

STUDY MATERIAL

Railway Study Material

Demo Edition - RRB NTPC - Group D - ALP - JE

Maths - Reasoning - General Science - GK

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About this demo

This is a **free demo PDF** from StudyElites. It contains sample chapters from our Railway Study Material so you can preview the structure, depth and quality of the full product before purchasing.

The full version (delivered after purchase via Razorpay) contains the complete chapter set, additional practice sets, previous-year question walkthroughs, and topic-wise quick-revision sheets. This demo covers approximately 15-20% of the full material.

What's inside this demo

- Chapter 1 - Mathematics (Number system, ratio, time & work)
- Chapter 2 - General Intelligence & Reasoning (Analogy, classification, series)
- Chapter 3 - General Science (Physics, Chemistry, Biology (10th level))
- Practice question bank with answer key

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Chapter 1

Mathematics

1.1 Number system, ratio, time & work

RRB Mathematics is 10th-standard level but time-bound. 30 questions in 30 minutes means 1 minute per question. Topics covered include number system, LCM/HCF, ratio & proportion, percentage, averages, time & work, time-speed-distance, and mensuration.

Time & Work problems are a recurring favourite. The trick is to use the 'unit work' method: if A can do a job in 12 days, A's 1-day work = $1/12$. Combined with B's rate, you can find the combined time easily.

1.2 Important Formulas / Rules

Work = rate x time

If A does job in 'a' days, A's 1-day work = $1/a$

Combined rate = $1/a + 1/b$ (work per day)

Speed = distance / time

1.3 Worked Example

Example. A does a job in 12 days, B in 18 days. Together, how many days?

A's rate = $1/12$, B's rate = $1/18$. Combined = $1/12 + 1/18 = 3/36 + 2/36 = 5/36$. Time = $36/5 = 7.2$ days.

Chapter 2

General Intelligence & Reasoning

2.1 Analogy, classification, series

RRB Reasoning is mostly non-verbal and verbal analogy, classification, series completion, coding-decoding, and blood relations. 25-30 questions in CBT (computer-based test). The key is rapid pattern recognition.

Coding-decoding uses letter shifting. If FRIEND is coded as GSJFOE (each letter +1), then HUSBAND will be coded by shifting each letter +1 as well. Memorise alphabet positions: A=1, B=2, ..., Z=26.

2.2 Important Formulas / Rules

Letter position: A=1, B=2, ..., Z=26

Reverse position: A=26, B=25, ..., Z=1

Direct coding: each letter shifted by +n

2.3 Worked Example

Example. If FRIEND = GSJFOE, how is HUSBAND coded?

Each letter shifted +1: H->I, U->V, S->T, B->C, A->B, N->O, D->E. Answer: IVTCBOE.

Chapter 3

General Science

3.1 Physics, Chemistry, Biology (10th level)

RRB General Science covers 10th-standard Physics, Chemistry and Biology. Topics: human body, plants, light/sound/electricity, acids/bases, periodic table basics. 20-25 questions appear in CBT.

Physics high-yield topics: Newton's laws of motion, light (reflection, refraction, lenses), electricity (Ohm's law $V=IR$), sound (frequency, amplitude). Chemistry: pH scale (acid < 7 , base > 7 , neutral = 7), common elements (H, O, C, N, Na, Cl, Fe). Biology: human digestive system, photosynthesis, cell structure.

3.2 Important Formulas / Rules

Ohm's Law: $V = IR$

Speed of light in vacuum: 3×10^8 m/s

Speed of sound in air: 343 m/s (at 20 deg C)

pH: <7 acid, $=7$ neutral, >7 base

3.3 Worked Example

Example. A 10 ohm resistor carries 2 amp current. Voltage across it?

Ohm's law: $V = IR = 2 \times 10 = 20$ volts.

Practice Question Bank

10 sample questions with answer key.

Maths

Q1. A does a job in 12 days, B in 18 days. Together, days?

- (A) 6
- (B) 7.2
- (C) 8.5
- (D) 9

Answer: (B) 7.2 - $\frac{1}{12} + \frac{1}{18} = \frac{5}{36}$; time = $\frac{36}{5} = 7.2$ days.

Q2. Speed of a 150m train crossing a pole in 7.5 sec?

- (A) 60 km/h
- (B) 72 km/h
- (C) 80 km/h
- (D) 90 km/h

Answer: (B) 72 km/h - speed = $\frac{150}{7.5} = 20$ m/s = 72 km/h.

Q3. If 3 men or 5 women do a job in 12 days, 1 man + 1 woman will take how many days?

- (A) 12
- (B) 15
- (C) 18
- (D) 20

Answer: (C) 18 - $3m = 5w \Rightarrow 1m = \frac{5}{3}w$. Combined = $1m + 1w = \frac{5}{3}w + 1w = \frac{8}{3}w$. Time = $(5w \times 12) / (\frac{8}{3}w) = 60 / \frac{8}{3} = 18$ days.

Reasoning

Q4. Complete: 2, 6, 12, 20, 30, ?

- (A) 36
- (B) 40
- (C) 42
- (D) 48

Answer: (C) 42 - differences 4, 6, 8, 10, 12; $30+12=42$.

Q5. FRIEND = GSJFOE, HUSBAND = ?

- (A) ITTCBOE
- (B) IVTCBOE
- (C) ITUCBOE
- (D) IVUCBOE

Answer: (A) ITTCBOE - each letter +1.

Science

Q6. Ohm's law: $V = ?$

- (A) IR
- (B) I/R
- (C) R/I
- (D) $I+R$

Answer: (A) $V = IR$.

Q7. pH of a neutral solution?

- (A) 0
- (B) 7
- (C) 14
- (D) 10

Answer: (B) 7 - neutral pH is 7.

Q8. Speed of light in vacuum?

- (A) 3×10^6 m/s
- (B) 3×10^8 m/s
- (C) 3×10^{10} m/s
- (D) 3×10^5 m/s

Answer: (B) 3×10^8 m/s.

GK

Q9. Indian Railways was founded in which year?

- (A) 1853
- (B) 1900
- (C) 1947
- (D) 1950

Answer: (A) 1853 - first train ran from Bombay to Thane on 16 April 1853.

Q10. Which is the longest railway platform in India?

- (A) Gorakhpur
- (B) Kollam
- (C) Hubli
- (D) Bilaspur

Answer: (C) Hubli - 1,505 metres (as of 2023).

About StudyElites & How to Get Full Access

This demo PDF was prepared by **StudyElites.online** as a free preview of our Railway Study Material. We hope it gave you a clear sense of the structure, depth and tone of our products.

What you get in the full version

The complete Railway Study Material includes 10+ chapters with full theory, 100+ worked examples, shortcut techniques, previous-year question walkthroughs and topic-wise practice sets with detailed solutions. The full PDF is roughly 150-250 pages.

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